

Phase 2

Molecular and Chemical Characterization of the Interstitial 17p13.3 Microdeletion

Abstract

An interstitial 17p13.3 microdeletion spanning GRCh37/hg19 chr17:1,411,408–1,841,103 removes approximately 429.7 kb and disrupts a minimum of 17 annotated loci. Phase 1 established that the deletion fully encompasses eleven protein-coding genes—PITPNA, SLC43A2, SCARF1, RILP, PRPF8, TLCD2, WDR81, SERPINF2, SERPINF1, SMYD4, and RPA1—while partially disrupting INPP5K and RTN4RL1. Exon-level analysis demonstrated that the partial genomic designations substantially underestimate coding disruption: approximately 49.6% of the INPP5K amino-acid-coding sequence and approximately 99.0% of the RTN4RL1 CCDS are deleted under the reported-coordinate model. Four additional noncoding loci—PITPNA-AS1, MIR22HG, MIR22, and RN7SL105P—are completely encompassed.

Phase 2 systematically characterized the molecular functions, biochemical mechanisms,

structural evidence, loss-of-function biology, heterozygous dosage evidence, neural relevance, retinal/visual relevance, and experimental tractability of all affected loci. The principal finding is not the identification of a single candidate disease gene. Instead, the interval contains several independently characterized regulators of membrane lipid composition, phosphoinositide metabolism, endosomal maturation, intracellular neuronal transport, extracellular neuronal signaling, retinal trophic support, RNA processing, post-transcriptional regulation, epigenetic modification, DNA replication and repair, amino-acid transport, efferocytosis, and fibrinolysis.

The strongest multi-gene convergence involves INPP5K, PITPNA, TLCD2, WDR81,

RILP, and SERPINF1, which affect different levels of membrane and phospholipid

homeostasis. RTN4RL1 extends this system into cell-surface neuronal structural signaling. PRPF8 provides an independent spliceosomal mechanism with definitive evidence for PRPF8-related autosomal-dominant retinopathy, although the established retinal disease mechanism cannot be assumed to be equivalent to whole-gene haploinsufficiency. RPA1 provides the strongest direct evidence that heterozygous loss within the interval can produce a measurable human cellular phenotype: RPA1-containing deletions have been associated experimentally with reduced RPA1 abundance and impaired ATR-dependent DNA-damage responses, including functional rescue following RPA1 restoration (Clinical Genome Resource, “PRPF8-Related Retinopathy”).

The noncoding analysis identifies the MIR22HG/MIR22 unit as the most consequential regulatory locus outside the protein-coding set. Complete Mir22 loss exacerbates experimental epilepsy in mice, demonstrating neurological functionality while not establishing a human heterozygous seizure phenotype (Silva et al.).

Collectively, Phase 2 supports a model of multisystem molecular dosage disruption **with a prominent membrane/phosphoinositide/endosomal convergence, but does not**

establish a definitive clinical phenotype for the specific heterozygous deletion. The principal unresolved question is whether one-copy loss produces sufficiently large, tissue-specific and developmentally coincident perturbations to alter retinal or neurological physiology. This question provides the basis for subsequent spatiotemporal functional-genomic and pathway analyses.

Keywords: 17p13.3; microdeletion; copy-number variation; phosphoinositides; endosome;

membrane homeostasis; PRPF8; RPA1; SERPINF1; WDR81; MIR22; dosage sensitivity; retinal biology; neurodevelopment.

1. Introduction

Copy-number variants can affect biological systems in ways that are not adequately represented by simple gene-counting approaches. A deletion may remove entire genes, disrupt only selected exons, eliminate regulatory RNAs, alter dosage of multimeric complexes, or simultaneously perturb several mechanistically related pathways. Consequently, interpretation of a rare CNV requires more than identification of disease-associated genes located within the interval.

The deletion evaluated in this project is an interstitial 17p13.3 microdeletion reported on the GRCh37/hg19 assembly as:

chr17:1,411,408–1,841,103

corresponding to a nominal span of:

429,696 bp, or approximately 429.7 kb.

Phase 1 established a minimum inventory of 17 affected loci. Thirteen are protein-coding genes:

1. INPP5K
2. PITPNA
3. SLC43A2
4. SCARF1
5. RILP
6. PRPF8
7. TLCD2
8. WDR81
9. SERPINF2
10. SERPINF1
11. SMYD4
12. RPA1
13. RTN4RL1

Four additional affected loci are noncoding:

14. PITPNA-AS1
15. MIR22HG
16. MIR22
17. RN7SL105P.

Eleven of the protein-coding genes are completely encompassed. INPP5K and RTN4RL1 lie at the CNV boundaries. Detailed coding analysis showed that INPP5K loses approximately 49.6% of its amino-acid-coding sequence, including a large portion of the catalytic domain, while RTN4RL1 loses approximately 99.0% of its CCDS. These findings demonstrate why genomic overlap percentage alone is an inadequate measure of functional gene disruption.

The purpose of Phase 2 was therefore to move from structural annotation to molecular **characterization. The central question was:**

What molecular functions are disrupted by heterozygous dosage loss of the affected loci, and which of these functions provide experimentally testable biochemical mechanisms for subsequent visual-system and neurological analysis?

The analysis was deliberately phenotype-neutral at the outset. Genes were not prioritized because they appeared to fit a proposed clinical phenotype. Instead, every affected protein-coding gene was characterized independently, followed by parallel evaluation of the noncoding loci.

The resulting analysis reveals a deletion that affects multiple independent cellular systems but also contains a notable concentration of genes involved in membrane lipid **regulation, phosphoinositide biology, endosomal maturation, intracellular trafficking, and neuronal membrane signaling. The distinction between these two observations—multisystem**

disruption and local mechanistic convergence—is central to interpretation.

2. Analytical Framework

2.1 Structural state

Each locus was first classified according to the Phase 1 structural model.

The eleven fully encompassed protein-coding genes were treated as complete heterozygous gene losses.

INPP5K was classified as a major partial coding disruption because approximately 666 of 1,344 amino-acid-coding nucleotides are deleted under the reported-coordinate model.

RTN4RL1 was classified as a functionally near-complete coding disruption because approximately 1,313 of 1,326 CCDS nucleotides are deleted.

All four noncoding loci are completely encompassed.

This classification describes the affected chromosome only. The homologous chromosome 17 retains the second genomic copy unless another independent variant affects it.

2.2 Evidence categories

Each protein-coding gene was evaluated independently across the following dimensions:

- structural disruption;
- biochemical characterization;
- experimental structural evidence;
- experimental loss-of-function evidence;
- human heterozygous dosage evidence;
- neural relevance;
- direct retinal/visual relevance;
- chemical and experimental tractability.

These dimensions were intentionally maintained separately.

A protein may have exceptionally well-characterized biochemical activity but little evidence that heterozygous loss affects humans. Conversely, a gene may have comparatively modest mechanistic characterization while possessing direct human dosage evidence.

The noncoding loci were assessed using a modified framework emphasizing:

- RNA class;
- experimentally demonstrated regulatory activity;
- perturbation evidence;
- heterozygous dosage evidence;
- neural relevance;

- visual relevance;
- experimental tractability.

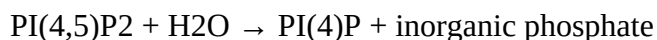
Protein-centered metrics such as catalytic domains, three-dimensional protein structures, pLI, and LOEUF were not mechanically transferred to lncRNAs, miRNAs, or pseudogenes.

3. Protein-Coding Locus Characterization

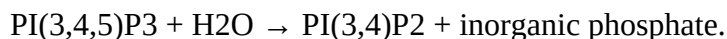
3.1 INPP5K: phosphoinositide 5-phosphate hydrolysis

INPP5K encodes an inositol polyphosphate 5-phosphatase. Its principal membrane substrates include PI(4,5)P₂ and PI(3,4,5)P₃.

The major reactions can be represented as:



and:



The reaction is Mg²⁺-dependent.

Phase 1 showed that the deletion removes approximately the first 666 coding nucleotides

of the canonical INPP5K coding sequence on the affected allele. Because INPP5K lies on the minus strand, this removes the 5' coding region. The deleted portion intersects the catalytic EEP domain and removes annotated catalytic residues, including residues involved in metal binding and proton-transfer chemistry.

A 2026 human INPP5K crystal structure resolved the catalytic domain at 1.9 Å in complex with the noncompetitive inhibitor CPD-1, establishing a structurally defined catalytic and allosteric system (Nomura et al.).

The deletion therefore cannot produce an intact canonical INPP5K catalytic domain from the affected allele under the reported-coordinate model. Whether the rearranged allele produces no transcript, an unstable transcript, or an aberrant protein remains unresolved.

INPP5K represents a strong biochemical candidate but weakly established human haploinsufficiency candidate.

3.2 PITPNA: phosphatidylinositol and phosphatidylcholine transfer

PITPNA encodes phosphatidylinositol transfer protein alpha, a soluble lipid-binding protein capable of transporting phosphatidylinositol and phosphatidylcholine between membrane environments.

PITPNA does not catalyze phosphate transfer or phospholipid hydrolysis. Its chemistry is instead based on reversible sequestration of a single phospholipid within a hydrophobic binding cavity.

Conceptually:

$\Rightarrow$

PI in membrane A + PITPNA

$\text{PITPNA} \cdot \text{PI} \rightarrow \text{PI in membrane B}$

with an analogous cycle for phosphatidylcholine.

PITPNA therefore affects the availability of phosphatidylinositol from which phosphorylated phosphoinositides are generated.

The neurological importance of PITPNA is strongly supported in the vibrator mouse. Reduction of PITPNA expression to approximately 20% of normal causes tremor, brainstem and spinal-cord neurodegeneration, and premature death (Hamilton et al.).

The index CNV completely deletes one PITPNA copy.

The important distinction is that the severe vibrator phenotype represents much greater reduction than an ordinary heterozygous deletion. Nonetheless, the model demonstrates that PITPNA possesses dose-dependent neurological biology.

3.3 SLC43A2: essential neutral amino-acid transport

SLC43A2 encodes LAT4, a plasma-membrane facilitated-diffusion transporter for neutral amino acids including:

- leucine;
- isoleucine;
- valine;
- methionine;
- phenylalanine.

Transport is not directly ATP driven and is largely sodium independent.

The protein therefore introduces a nutrient-transport mechanism distinct from the membrane-signaling cluster.

Potential downstream consequences of altered SLC43A2 activity include changes in:

- essential amino-acid availability;
- leucine-sensitive mTORC1 signaling;
- methionine-dependent S-adenosylmethionine metabolism;
- general cellular anabolic state.

However, these are downstream possibilities rather than established consequences of heterozygous SLC43A2 loss.

Experimental loss-of-function models establish important placental, intestinal and renal transport roles, but current evidence for a direct heterozygous neurological or retinal phenotype is weak.

SLC43A2 should therefore remain an independent nutrient-transport axis rather than being forced into the principal neural convergence model.

3.4 SCARF1: scavenger recognition and efferocytosis

SCARF1 encodes a type-I scavenger receptor involved in recognition and removal of apoptotic material.

One important mechanism involves C1q acting as a molecular bridge between phosphatidylserine-rich apoptotic membranes and SCARF1.

The functional sequence is approximately:

apoptotic membrane

↓

C1q recognition

↓

SCARF1 engagement

↓

efferocytosis.

SCARF1-deficient mice develop defective apoptotic-cell clearance and lupus-like autoimmunity, establishing strong loss-of-function biology.

However, SCARF1 currently contributes little direct evidence to the visual-neurological hypothesis.

Its principal Phase 2 significance is the demonstration that the deletion also affects an immune-clearance and damage-recognition pathway.

3.5 RILP: RAB7-dependent late-endosomal transport

RILP encodes Rab-interacting lysosomal protein, an effector of active RAB7A.

The relevant interaction is:

$\Rightarrow$

RAB7A-GTP + RILP

RAB7A-GTP:RILP complex.

RILP does not hydrolyze GTP. Instead, it recognizes the active GTP-bound state of RAB7A and connects late endosomes and lysosomes to:

- dynein;
- dynactin;
- HOPS;
- additional trafficking machinery.

This enables minus-end-directed intracellular transport and perinuclear positioning of late endosomal compartments.

Recent neuronal work has strengthened the relevance of this system substantially. Interference with RILP-RAB7A-mediated late-endosome transport impairs dendritic cargo movement and dendrite arborization (Yap et al.).

RILP therefore provides direct mechanistic evidence that the deletion affects neuronal intracellular trafficking, although human haploinsufficiency remains weakly characterized.

3.6 PRPF8: spliceosomal RNA processing

PRPF8 encodes one of the central structural and regulatory proteins of the spliceosome.

It contacts pre-mRNA, U5 snRNA, U6 snRNA, and multiple spliceosomal proteins and contributes to organization of the catalytic center.

The two principal pre-mRNA splicing reactions are transesterification reactions.

First:

branch-point adenosine 2'-OH attacks the 5' splice site
forming the intron lariat.

Second:

the released exon 1 3'-OH attacks the 3' splice site
joining the two exons.

PRPF8 itself is not accurately described as a conventional phosphodiesterase or ligase.

Catalysis is largely RNA based, with PRPF8 organizing the molecular environment in which the reactions occur.

The index deletion removes one complete PRPF8 copy.

PRPF8 is exceptional because its retinal relevance is established independently of this CNV. ClinGen classifies autosomal-dominant PRPF8-related retinopathy as Definitive (Clinical Genome Resource, "PRPF8-Related Retinopathy").

Recent experimental work on retinitis-pigmentosa-associated PRPF8 variants showed reduced pre-mRNA binding, slower intron splicing and altered expression of hundreds of genes in retinal pigment epithelial cells (Zimmann et al.).

Heterozygous PRPF8 variants have also been reported in individuals with neurodevelopmental disorders (O'Grady et al.).

A critical limitation remains: classic PRPF8-associated retinopathy is usually linked to specific pathogenic variants, particularly C-terminal variants, and is not established as a simple whole-gene haploinsufficiency disorder.

Consequently:

PRPF8 has definitive retinal disease relevance, but the phenotype associated specifically with heterozygous whole-gene deletion remains incompletely defined.

3.7 TLCD2: phospholipid acyl-chain composition and membrane fluidity

TLCD2 is a small multi-pass membrane protein belonging to the TLC-domain family.

Its experimentally supported function involves regulation of which fatty-acid species are incorporated into membrane phospholipids.

Reduction of TLCD1/2 activity increases incorporation of long-chain polyunsaturated fatty acids into phospholipids and alters membrane fluidity (Ruiz et al.).

The mechanism is not equivalent to fatty-acid synthesis or bulk uptake. TLCD proteins appear to influence intracellular phospholipid composition, particularly the balance among saturated, monounsaturated and polyunsaturated fatty-acid-containing phospholipids.

This is important because membrane acyl-chain composition affects:

- fluidity;
- curvature;
- protein mobility;
- organelle interactions;
- susceptibility to lipid peroxidation.

The index deletion removes one complete TLCD2 copy.

Human haploinsufficiency is not established, and TLCD1 may provide substantial functional redundancy.

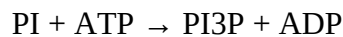
TLCD2 is therefore a strong biochemical membrane candidate but weak dosage candidate.

3.8 WDR81: PI3P regulation and endosome maturation

WDR81 is a large endosomal regulatory protein with particularly important consequences for the emerging membrane model.

WDR81, together with WDR91, negatively regulates the class III phosphatidylinositol 3kinase complex involved in formation of phosphatidylinositol-3-phosphate.

The relevant enzymatic conversion is:



performed by VPS34/PIK3C3.

WDR81 does not catalyze the reaction. It restrains the PI3K complex.

Loss of WDR81/WDR91 activity therefore produces:

excess PI3P

↓

excessive early-endosome fusion

↓

defective early-to-late endosome conversion

↓

abnormal receptor trafficking.

This regulatory mechanism has been directly demonstrated experimentally (Liu et al.).

WDR81 loss also disrupts EGFR trafficking and MAPK signaling in neural progenitor cells and reduces progenitor proliferation and brain growth in experimental models (Carpentieri et al.).

WDR81 additionally participates in selective autophagic removal of ubiquitinated protein aggregates.

Thus WDR81 links:

phosphoinositide chemistry

to:

endosomal identity

to:

neural developmental signaling.

Established human WDR81 disorders are predominantly recessive, so heterozygous dosage consequences remain much less certain.

3.9 SERPINF2: plasmin inhibition and fibrinolysis

SERPINF2 encodes alpha-2-antiplasmin, the principal physiological inhibitor of plasmin.

Plasmin degrades fibrin during fibrinolysis.

Alpha-2-antiplasmin inhibits plasmin using the classical serpin suicide-inhibition mechanism:

plasmin + alpha-2-antiplasmin

↓

reactive-center-loop cleavage

↓

covalent serpin-protease intermediate

↓

conformational trapping

↓

inactive plasmin-antiplasmin complex.

SERPINF2 is additionally crosslinked into fibrin by factor XIIIa, placing the inhibitor directly within the clot.

Severe biallelic deficiency causes hyperfibrinolytic bleeding.

Importantly, heterozygous carriers can demonstrate approximately half-normal biochemical activity and may range from asymptomatic to clinically significant bleeding following surgery, trauma or dental procedures. A review of 123 reported cases confirms this variable heterozygous phenotype (Matrane et al.).

SERPINF2 therefore provides meaningful evidence that one-copy loss can produce a **measurable biochemical dosage effect, although it does not materially strengthen the principal**

visual-neural model.

3.10 SERPINF1: PEDF neurotrophic and retinal signaling

SERPINF1 encodes pigment epithelium-derived factor, or PEDF.

Despite being a serpin-family protein, PEDF is not a conventional protease inhibitor. Its serpin fold has instead been adapted to extracellular signaling.

PEDF has established:

- neurotrophic activity;
- antiangiogenic activity;
- extracellular-matrix binding;

- retinal protective effects;
- roles in bone mineralization.

One important receptor is PNPLA2/PEDF-R, which possesses phospholipase activity.

PEDF therefore connects an extracellular trophic signal to membrane-lipid metabolism.

Experimental ablation of PEDF-R in mice causes altered retinal phospholipid composition, photoreceptor abnormalities, apoptosis and attenuation of electroretinographic responses. Even heterozygous *Pnpla2* animals demonstrated electrophysiological attenuation in that experimental system (Bernardo-Colón et al.).

Heterozygous human SERPINF1 mutation carriers provide especially useful dosage evidence. Eighteen carriers showed reduced SERPINF1 transcript levels and significantly reduced circulating PEDF but no detectable abnormalities in bone density or related skeletal parameters (Al-Jallad et al.).

This demonstrates:

heterozygous loss

↓

measurable biochemical dosage reduction

without necessarily producing:

a penetrant clinical phenotype.

SERPINF1 is therefore one of the strongest direct retinal-biological candidates in the interval, while remaining an incompletely established heterozygous retinal disease candidate.

3.11 SMYD4: protein lysine methylation and epigenetic regulation

SMYD4 encodes a SET- and MYND-domain lysine methyltransferase.

The fundamental methyltransferase reaction is:

protein lysine + S-adenosylmethionine

↓

methylated lysine + S-adenosylhomocysteine.

SMYD4 therefore directly performs a covalent chemical modification.

Experimental evidence implicates SMYD4 in:

- H3K4 methylation;
- interaction with HDAC1;
- regulation of histone acetylation;
- developmental transcription.

More recent human-cell work has demonstrated nonhistone activity. SMYD4 promotes lysine monomethylation of MYH9, increasing its ubiquitination and degradation and thereby influencing WNT/beta-catenin signaling (Yang et al.).

Despite this increasingly strong biochemical characterization, human heterozygous dosage sensitivity remains weak.

SMYD4 is therefore best treated as an epigenetic and post-translational regulatory component rather than a major direct neural or retinal candidate.

3.12 RPA1: single-stranded DNA protection and ATR signaling

RPA1 encodes the 70-kDa subunit of the heterotrimeric Replication Protein A complex.

RPA binds exposed single-stranded DNA during:

- replication;
- stalled replication forks;
- homologous recombination;
- DNA repair;

- telomere replication.

Binding is noncovalent and mediated through multiple OB-fold DNA-binding domains.

RPA-coated ssDNA also recruits ATRIP and contributes to activation of the ATR DNA damage checkpoint.

The index deletion completely removes one RPA1 copy.

RPA1 is particularly important because human cellular haploinsufficiency has already been experimentally demonstrated in the context of contiguous gene deletions. Patient-derived cells containing heterozygous RPA1 deletions demonstrated impaired ATR-dependent DNA damage responses (O'Driscoll et al.).

The original experiments further demonstrated correction after RPA1 restoration, strengthening attribution of the cellular phenotype to RPA1 dosage.

Current ClinGen dosage sensitivity remains formally Awaiting Review, emphasizing that a molecular haploinsufficiency phenotype does not automatically constitute a fully curated clinical haploinsufficiency syndrome (Clinical Genome Resource, "RPA1 Dosage Sensitivity Curation").

RPA1 therefore provides the strongest evidence in the interval that:

heterozygous gene loss itself can produce a directly measurable molecular

defect.

Its strongest consequence is genome-maintenance vulnerability rather than established visual-system dysfunction.

3.13 RTN4RL1: neuronal extracellular signaling and axonal plasticity

RTN4RL1 encodes NgR3, a GPI-anchored neuronal receptor.

Although only a small percentage of the genomic RTN4RL1 span overlaps the deletion, approximately 99% of the CCDS is removed under the Phase 1 model.

The affected allele therefore cannot encode an intact canonical NgR3 receptor.

NgR3 contains extracellular leucine-rich repeats and recognizes glycosaminoglycan components of chondroitin sulfate proteoglycans.

NgR1 and NgR3 jointly mediate CSPG-associated restriction of axonal growth. Combined receptor loss enhances regeneration following optic-nerve injury, whereas isolated single-receptor loss produces substantially weaker effects (Dickendesher et al.).

This establishes two important points.

First, RTN4RL1 is directly relevant to neuronal structural signaling.

Second, substantial Nogo-receptor-family redundancy exists.

RTN4RL1 therefore provides strong neural and direct visual-pathway experimental relevance, but relatively weak evidence that heterozygous loss alone produces a human phenotype.

4. Noncoding Loci

4.1 PITPNA-AS1

PITPNA-AS1 is an antisense long noncoding RNA.

Experimental cancer-cell studies demonstrate that it can participate in:

- miRNA sequestration;
- RNA-protein interactions;
- mRNA stabilization;
- regulation of EGFR/PI3K-AKT and other signaling pathways.

However, effects are strongly context dependent, and evidence remains dominated by cancer models.

Critically, there is insufficient evidence to assume that PITPNA-AS1 loss directly reduces PITPNA expression merely because the genes are antisense neighbors.

PITPNA-AS1 should therefore remain in transcriptomic analysis but currently represents a low-to-moderate-priority regulatory candidate.

4.2 MIR22HG

MIR22HG is a long noncoding host transcript containing the MIR22 precursor.

This creates two molecular layers:

MIR22HG lncRNA function

and:

miR-22 production.

The host RNA has experimentally demonstrated regulatory functions independent of mature miR-22, including RNA-protein interactions and effects on major signaling pathways.

The direction of MIR22HG activity is highly context dependent, particularly across different malignancies.

The index deletion removes one MIR22HG copy and simultaneously removes one genomic MIR22 precursor.

This structural relationship is important:

MIR22HG and MIR22 are distinct functional annotations but not

independent deletion events.

4.3 MIR22

MIR22 encodes a regulatory microRNA processed into mature miR-22 species that associate with Argonaute-containing RISC complexes.

The simplified mechanism is:

miR-22 + Argonaute

↓

target-mRNA base pairing

↓

translation suppression and/or increased target-RNA degradation.

MicroRNAs are capable of regulating many transcripts simultaneously.

Consequently, MIR22 loss has the potential to affect regulatory networks rather than a single biochemical reaction.

Direct neurological evidence exists.

Genetic deletion of Mir22 in mice resulted in earlier onset, increased frequency and increased duration of spontaneous seizures following experimental status epilepticus (Silva et al.).

Importantly, baseline brain structure and behavior were comparatively normal in the knockout model before the epileptogenic challenge.

This suggests that miR-22 may influence response or resilience to neural stress rather than being required for basic nervous-system formation.

The finding does not establish that one-copy MIR22 loss causes seizures in humans.

MIR22 nevertheless becomes the most compelling noncoding neurological candidate in the interval.

4.4 RN7SL105P

RN7SL105P is an inferred 7SL-related pseudogene.

Functional 7SL RNA is an important component of the signal recognition particle, but this ancestral relationship cannot be transferred automatically to RN7SL105P.

No convincing locus-specific evidence currently establishes:

- functional RNA production;
- SRP participation;
- neural activity;
- retinal activity;
- clinically significant dosage sensitivity.

RN7SL105P is therefore retained for structural completeness but presently receives no major mechanistic weight.

5. Integrated Molecular Characterization Matrix

Principal

Structural

Heterozygous

Locus

Neural relevance

Visual relevance

molecular system

consequence

dosage evidence

~49.6% coding loss;

Phosphoinositide INPP5K

catalytic region

Weak

Moderate

Limited–moderate

hydrolysis

disrupted PITPNA

PI/PC transfer

Full gene loss

Moderate

Strong

Limited

Essential amino-acid SLC43A2

Full gene loss

Weak

Limited

Minimal

transport

Efferocytosis/ SCARF1

Full gene loss

Weak

Limited

Minimal

immune clearance

Late-endosome RILP

Full gene loss

Weak

Strong

Limited

transport

Moderate/uncertain PRPF8

Pre-mRNA splicing

Full gene loss

Strong

Very strong

mechanism

Membrane TLCD2

phospholipid

Full gene loss

Weak

Limited

Minimal

composition

PI3P/endosome WDR81

Full gene loss

Weak

Strong

Limited

maturation

Fibrinolysis/plasmin

Moderate–strong SERPINF2

Full gene loss

Limited

Minimal

inhibition

biochemical

PEDF trophic/retinal

Moderate SERPINF1

Full gene loss

Strong

Very strong

lipid signaling

biochemical

Protein SMYD4

methylation/epigenet

Full gene loss

Weak

Limited

Minimal

ic regulation

DNA

Strongest direct RPA1

replication/repair/A

Full gene loss

Moderate

Minimal

molecular evidence

TR response

Axon/synapse RTN4RL1

~99% CCDS loss

Weak-moderate

Strong

Moderate

signaling PTPNA-AS1

lncRNA regulation

Full locus loss

Unknown

Limited

None established

lncRNA/miRNA- MIR22HG

Full locus loss

Unknown

Limited–moderate

Limited

host regulation

miRNA post-

Moderate–strong MIR22

transcriptional

Full locus loss

Unknown in humans

Limited

experimental

regulation

7SL-related RN7SL105P

Full locus loss

None established

None

None

pseudogene

6. Principal Mechanistic Convergence: Membrane, Phosphoinositide and Endosomal Homeostasis

The strongest Phase 2 finding is the presence of several independently characterized genes affecting different layers of membrane biology.

This cluster consists principally of:

PITPNA

→ phosphatidylinositol availability and lipid transfer

INPP5K

→ phosphoinositide 5-phosphate hydrolysis

WDR81

→ regulation of endosomal PI3P production

TLCD2

→ phospholipid fatty-acyl composition and membrane fluidity

SERPINF1

→ extracellular trophic signaling linked to phospholipid metabolism

RILP

→ RAB7-dependent late-endosomal transport.

These genes do not perform the same biochemical reaction.

Their importance lies in their positions at distinct regulatory layers of a common cellular

These genes do not perform the same biochemical reaction. Their importance lies in their positions at distinct regulatory layers of a common cellular system.

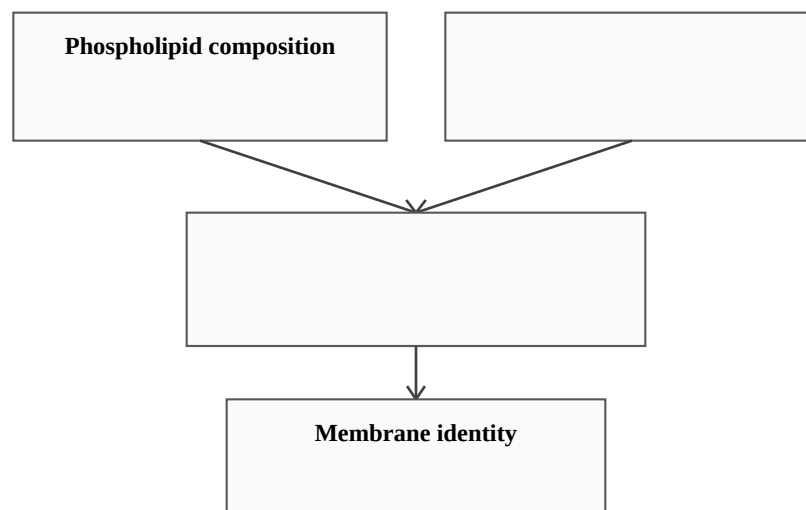
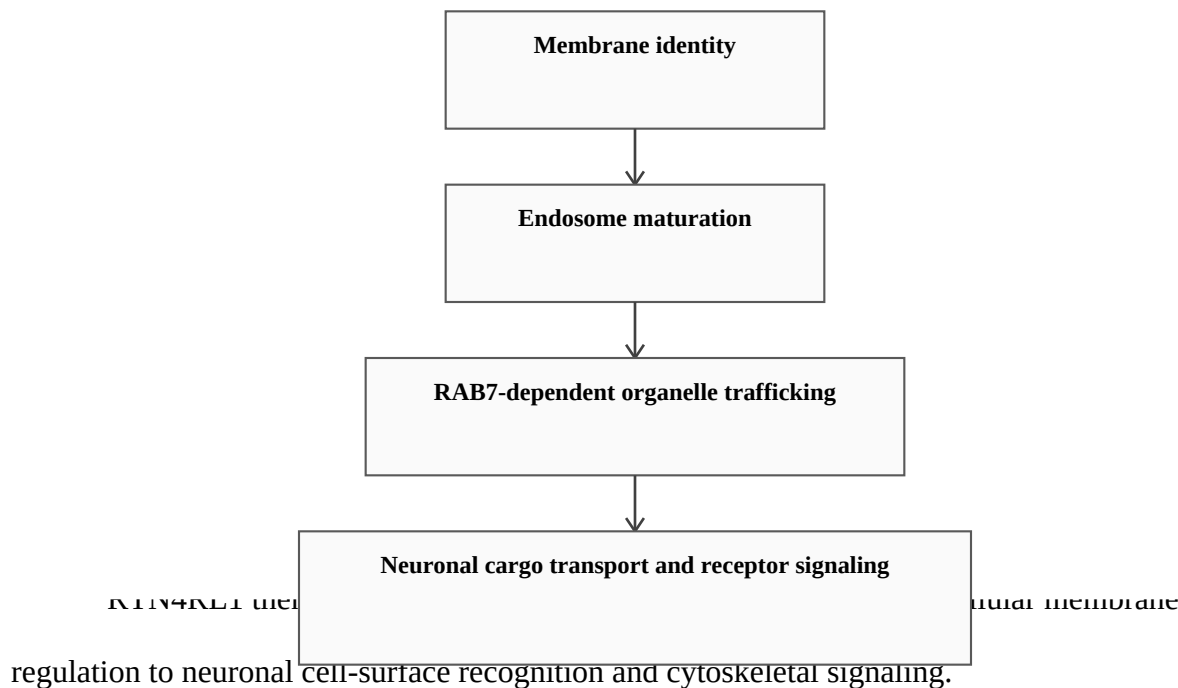
Simplified integrated model

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7. The Convergence Is Not Necessarily Directionally Additive

The observation that multiple genes affect the same broad system does not mean their deletion drives every molecular variable in the same direction.

This is particularly important for phosphoinositides.

Reduced PITPNA could potentially decrease efficient phosphatidylinositol supply.

Reduced WDR81 could increase endosomal PI3P because normal WDR81 restrains class III PI3K activity.

Reduced INPP5K could impair degradation of PI(4,5)P₂ and PI(3,4,5)P₃.

These effects occur:

- on different phosphoinositide species;
- in different cellular compartments;
- through different reactions.

Consequently, the current evidence supports:

dysregulated phosphoinositide homeostasis

rather than:

uniformly increased or uniformly decreased phosphoinositide signaling.

The possibility of opposing and compensatory effects is one of the most important reasons that transcriptomic and pathway analysis must precede causal interpretation.

8. Independent Regulatory Axes

8.1 RNA processing

PRPF8 creates a potentially broad regulatory mechanism because altered spliceosomal function can affect many transcripts simultaneously.

A later testable question is whether PRPF8 dosage perturbation preferentially affects transcripts involved in:

- phosphoinositide metabolism;
- endosomal trafficking;
- neuronal signaling;
- retinal maintenance.

Such an observation would connect PRPF8 to the membrane cluster through functional convergence.

It cannot currently be assumed.

8.2 Post-transcriptional RNA regulation

MIR22HG/MIR22 introduces a second RNA regulatory layer.

Unlike PRPF8, which regulates intron removal, miR-22 controls mRNA abundance and translation through sequence-specific RISC targeting.

Thus the deletion simultaneously affects:

pre-mRNA processing

and:

post-transcriptional gene regulation.

MIR22's experimental influence on epileptogenesis makes this mechanism relevant to later neurological analysis (Silva et al.).

8.3 Epigenetic regulation

SMYD4 contributes covalent protein methylation and chromatin-associated regulation.

This introduces a third level of information control:

chromatin/protein modification

↓

RNA processing

↓

post-transcriptional miRNA regulation.

Whether these three regulatory systems converge on common target pathways remains unknown.

8.4 Genome maintenance

RPA1 establishes a separate genome-integrity axis.

The deletion therefore affects not only membrane signaling and RNA processing but also:

- replication-fork protection;
- ssDNA stabilization;
- ATR checkpoint activation;
- homologous recombination;
- telomere-associated DNA metabolism.

The direct human cellular haploinsufficiency evidence for RPA1 makes this mechanism especially important even though it is not primarily visual (O'Driscoll et al.).

9. Additional Systemic Axes

9.1 Amino-acid transport

SLC43A2 contributes an essential-amino-acid transport mechanism.

This potentially affects nutrient sensing and metabolism but currently has limited direct evidence for visual-neural involvement.

9.2 Immune clearance

SCARF1 contributes apoptotic-cell clearance and immune homeostasis.

Its inclusion demonstrates that the deletion can influence extracellular damage recognition and immune regulation independently of the neural mechanisms.

9.3 Hemostasis

SERPINF2 contributes fibrinolytic regulation.

Its heterozygous carrier literature is important because it demonstrates that reduced gene dosage can result in measurable biochemical loss of function even when clinical penetrance is incomplete (Matrane et al.).

10. Heterozygous Dosage Is the Principal Unresolved Variable

The most important lesson from Phase 2 is that gene importance and haploinsufficiency are not equivalent.

For most loci, severe experimental loss establishes biological importance.

It does not establish that reducing gene copy number from two to one produces a clinically significant effect.

Three loci provide particularly useful dosage examples.

RPA1

Heterozygous genomic loss has been associated directly with:

- reduced RPA1 abundance;
- impaired ATR signaling;
- replication-stress abnormalities;
- molecular rescue after RPA1 restoration.

This constitutes the strongest direct one-copy molecular evidence in the interval (O'Driscoll et al.).

SERPINF1

Heterozygous carriers demonstrate reduced PEDF but no clearly penetrant skeletal phenotype.

This demonstrates:

molecular dosage effect without corresponding overt clinical disease. (Al-

Jallad et al.)

SERPINF2

Heterozygous carriers can show approximately half-normal biochemical activity, but clinical expression ranges from absent to substantial bleeding under physiological challenge (Matrane et al.).

These examples argue against a simplistic model in which:

one deleted allele = exactly 50% function = predictable disease.

Biological output depends on:

- transcriptional compensation;
- protein stability;

- pathway reserve;
- paralogous proteins;
- complex stoichiometry;
- tissue expression;
- developmental stage;
- environmental or physiological stress.

A more defensible concept is:

reduced functional reserve.

The deletion may reduce the capacity of multiple systems without eliminating them.

11. Neural and Visual-System Evidence

Phase 2 identifies several genes with strong neural relevance but only a smaller subset

with direct retinal or visual-system evidence.

11.1 PRPF8

PRPF8 provides the strongest established human retinal disease relationship.

ClinGen currently classifies PRPF8-related autosomal-dominant retinopathy as Definitive (Clinical Genome Resource, “PRPF8-Related Retinopathy”).

The relevant limitation is mechanism: known pathogenic retinal variants cannot automatically be equated with whole-gene deletion.

11.2 SERPINF1

SERPINF1 provides strong direct retinal biology through:

- PEDF neurotrophic signaling;
- photoreceptor survival;
- retinal ganglion-cell support;

- angiogenesis regulation;
- retinal phospholipid metabolism.

PEDF-R deficiency alters both retinal lipid composition and electrophysiological function experimentally (Bernardo-Colón et al.).

SERPINF1 is therefore one of the strongest mechanistic retinal candidates even though heterozygous SERPINF1 deletion is not an established dominant retinal disorder.

11.3 RTN4RL1

RTN4RL1 provides direct visual-pathway evidence through retinal ganglion-cell axon biology and optic-nerve regeneration.

Combined NgR1/NgR3 loss enhances regeneration after optic-nerve crush (Dickendesher et al.).

This supports an optic-pathway mechanism but not a normal visual-perception phenotype.

11.4 WDR81, RILP and PITPNA

These genes provide strong neuronal mechanisms involving:

- neural-progenitor proliferation;
- endosomal maturation;
- dendritic cargo transport;
- neuronal survival.

WDR81 loss alters endosomal EGFR trafficking and neural-progenitor proliferation (Carpentieri et al.).

RILP-RAB7A late-endosomal transport is directly relevant to dendrite arborization (Yap et al.).

PITPNA reduction produces severe neurodegeneration in mouse models (Hamilton et al.).

These findings justify neurological investigation but do not establish a specific human visual phenotype.

11.5 MIR22

MIR22 adds a neural regulatory mechanism associated experimentally with response to epileptogenic stress.

Complete knockout exacerbates spontaneous seizure development after status epilepticus (Silva et al.).

This evidence is mechanistically important but cannot establish seizure susceptibility from heterozygous MIR22 loss.

12. What Phase 2 Supports

The combined evidence supports several conclusions with relatively high confidence.

First, the deletion removes or severely disrupts multiple biologically functional loci.

Second, several affected genes independently regulate components of the same broader membrane/endosomal system.

Third, the convergence is chemically substantive rather than merely semantic. The affected functions include:

- lipid binding and transfer;
- phosphate-ester hydrolysis;
- PI3P synthesis regulation;
- phospholipid acyl-chain composition;
- phospholipase-linked extracellular signaling;

- RAB7-dependent organelle transport.

Fourth, several affected genes have direct experimental neural or retinal functions.

Fifth, heterozygous molecular dosage effects are demonstrable for at least some genes, most clearly RPA1 and biochemically for SERPINF1 and SERPINF2.

Sixth, the deletion also affects independent systems that should not be forced into the primary neural model.

The most defensible overall characterization is therefore:

a multisystem heterozygous dosage lesion containing a prominent

membrane/phosphoinositide/endosomal-neuronal convergence.

13. What Phase 2 Does Not Establish

Phase 2 does not establish that the deletion causes a specific clinical phenotype.

It does not demonstrate that all affected genes experience exactly 50% reduction in transcript or protein abundance.

It does not demonstrate that all affected pathways are impaired in the same direction.

It does not establish synergy or epistasis among deleted genes.

It does not establish that the membrane/endosomal convergence is active at pathological magnitude in retinal or neural tissue.

It does not establish that PRPF8 whole-gene deletion produces the same phenotype as established PRPF8 retinal variants.

It does not establish that heterozygous WDR81 loss reproduces recessive WDR81 disorders.

It does not establish that heterozygous SERPINF1 loss causes retinal degeneration.

It does not establish that heterozygous MIR22 loss causes seizures.

It does not establish:

- photophobia;
- pattern glare;
- visual stress;
- magnocellular dysfunction;
- cortical visual hyperexcitability;
- epilepsy;
- hearing abnormalities;
- congenital cardiac disease

as consequences of this deletion.

These remain hypotheses or phenotype variables for later investigation.

14. Limitations

14.1 CNV breakpoint resolution

The reported array coordinates may represent probe-defined boundaries rather than nucleotide-exact physical breakpoints.

Coding overlap is therefore established at CNV resolution, but exact breakpoint-junction biology remains unknown.

14.2 Transcript fate

For the partially deleted genes INPP5K and RTN4RL1, genomic disruption does not establish the exact RNA product.

Potential outcomes include:

- reduced transcription;
- aberrant splicing;

- unstable RNA;
- nonsense-mediated decay;
- unusual truncated products.

These require direct transcript evidence.

14.3 Dosage compensation

Complete deletion of one genomic allele does not prove a 50% reduction in cellular protein.

The intact allele may:

- compensate transcriptionally;
- fail to compensate;
- overcompensate;
- show tissue-specific regulation.

14.4 Paralogous redundancy

Several genes have functionally related family members.

Examples include:

RTN4RL1

with RTN4R and RTN4RL2,

and:

TLCD2

with TLCD1.

Such redundancy can reduce the phenotypic impact of heterozygous loss.

14.5 Evidence derived from severe loss

Many mechanistic studies use:

- homozygous knockout;
- severe hypomorphic alleles;
- RNA knockdown;
- cancer-cell overexpression;
- experimental injury.

These conditions cannot automatically be extrapolated to constitutional heterozygous deletion.

14.6 Tissue specificity

A molecular effect observed in:

- liver;
- kidney;
- tumor cells;
- fibroblasts;
- cultured neurons

may not occur identically in:

- retina;
- retinal pigment epithelium;
- retinal ganglion cells;
- thalamic visual nuclei;
- visual cortex.

This is the principal reason Phase 3 requires tissue- and cell-type-specific expression analysis.

14.7 Network inference

Pathway overlap does not establish direct interaction.

The terms:

- functional convergence;
- pathway overlap;
- network proximity

are currently appropriate.

Terms such as:

- synergistic interaction;
- epistasis;
- causal pathway

require additional evidence.

15. Phase 2 Working Model

The integrated molecular model can be represented as follows:

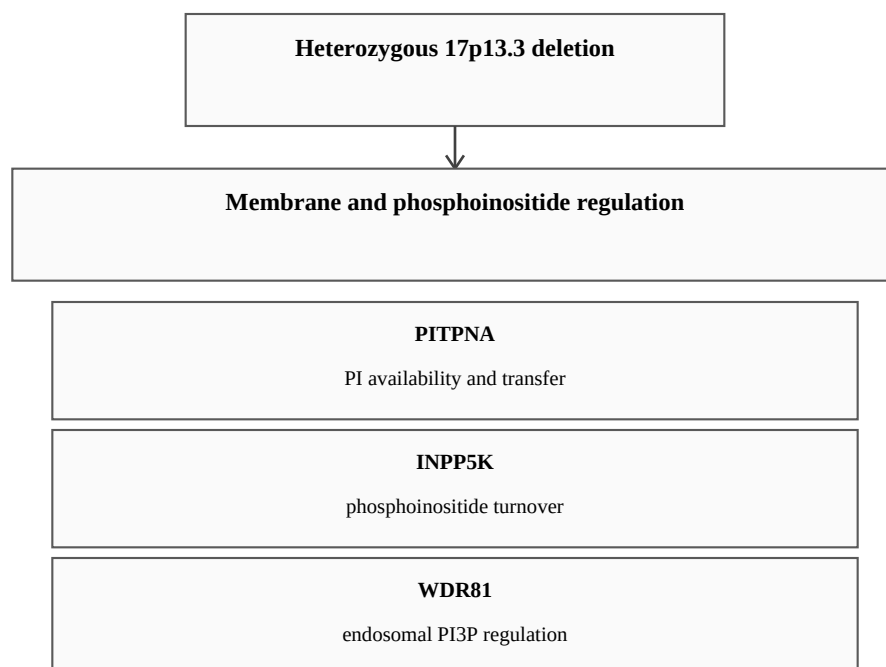
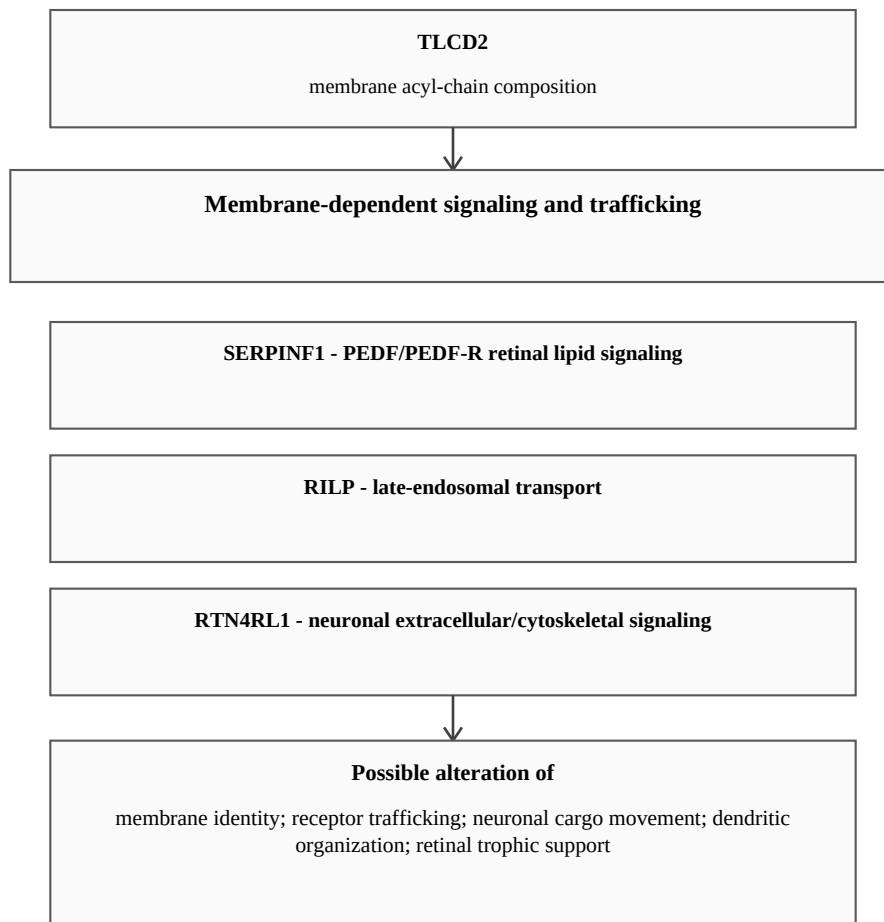
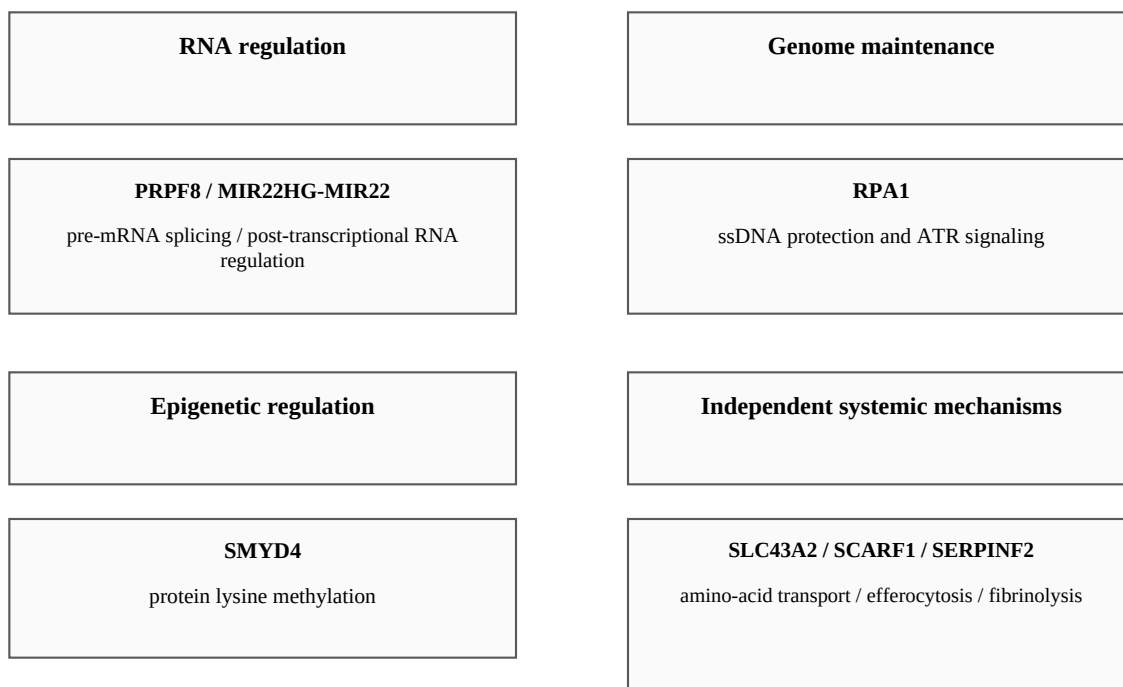


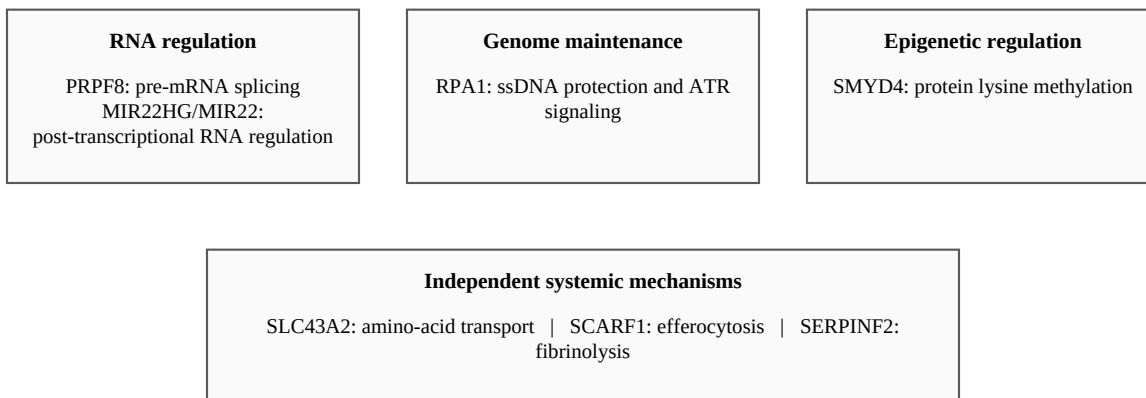
Figure continues on the next page.



In parallel with:



Phase 2 working model - independent parallel axes



16. Implications for the Next Analytical Phase

Phase 2 has reduced the research problem substantially.

The next question is no longer:

Which genes are deleted?

Nor is it simply:

Which deleted genes have neurological annotations?

The appropriate next question is:

Are the molecularly strongest candidate genes actually expressed together in the relevant human retinal and neurological cell populations, and do their expression patterns overlap at developmentally relevant times?

Phase 3 should therefore test:

1. tissue expression of all affected genes;
2. retinal cell-type expression;
3. neuronal cell-type expression;
4. developmental timing;
5. coexpression among the membrane/endosomal candidates;
6. whether PRPF8, MIR22 and other regulatory loci intersect those same expression programs;

7. whether candidate genes are expressed within visual system pathways strongly enough to make the Phase 2 convergence biologically relevant in vivo.

The analysis should remain genome-first rather than phenotype-first.

17. Conclusion

Phase 2 demonstrates that the chr17:1,411,408–1,841,103 microdeletion is molecularly complex.

The deletion does not merely remove a collection of unrelated genes. It simultaneously eliminates or severely disrupts multiple regulators of:

- phosphoinositide metabolism;
- phospholipid transport;
- membrane lipid composition;
- endosomal identity;

- intracellular trafficking;
- retinal trophic signaling;
- neuronal structural plasticity;
- RNA splicing;
- post-transcriptional RNA regulation;
- protein methylation;
- DNA replication and repair;
- amino-acid transport;
- immune clearance;
- fibrinolysis.

The strongest mechanistic convergence involves INPP5K, PITPNA, TLCD2, WDR81, RILP and SERPINF1, with RTN4RL1 extending the system into neuronal surface signaling.

PRPF8 independently provides definitive evidence for a retinal disease relationship under established pathogenic variant mechanisms, while RPA1 provides the strongest direct evidence that heterozygous genomic loss within the interval can produce a measurable human molecular defect (Clinical Genome Resource, “PRPF8-Related Retinopathy”; O’Driscoll et al.).

The MIR22HG/MIR22 unit contributes an additional regulatory mechanism with experimentally established effects on neural response to epileptogenic stress (Silva et al.).

The central conclusion is therefore not that this deletion has been assigned a phenotype.

Rather:

The deletion produces a credible multi-gene molecular architecture in which reduced functional reserve across membrane, endosomal, retinal, RNA-

regulatory and genome-maintenance systems is biologically plausible.

The magnitude, tissue specificity, developmental timing and clinical consequences of those reductions remain unresolved.

Accordingly, Phase 2 establishes mechanistic plausibility and research priorities, not causality.

The results provide a sufficiently developed molecular foundation to proceed to spatiotemporal functional genomics, where the principal hypothesis can be tested without assuming the clinical conclusion in advance.